

Arnav Srikanth

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EDUCATION

Carnegie Mellon University, Pittsburgh, PA

Master of Science in Mechanical Engineering - Research

May 2027

MetaMobility Lab, NASA Lunabotics Challenge

Current Courses: Mechanics of Manipulation, Advanced Mechatronic Design, Machine Learning and Artificial Intelligence for Engineers, Fundamentals of Robot Behavior Learning

University of Illinois Urbana-Champaign, Champaign, IL

Bachelor of Science in Mechanical Engineering

May 2022

Midwestern Robotics Design Competition, iRobotics Robobrawl, Illini Esports Community Events Chair

SKILLS

Applications: SolidWorks/SolidWorks Simulate, PTC Creo/Creo Mold Flow/Windchill, Fusion360, FEA

Manufacturing: 3D printing, CNC/manual milling, laser cutting, rapid prototyping, injection molding, ASME Y14.5

Programming: Python, MATLAB, C

WORK EXPERIENCE

Graduate Student Researcher | MetaMobility Lab, Pittsburgh, PA

August 2025 - Present

- Analyzed impacts of 3 distinct hip exoskeleton control approaches on 10 subjects, demonstrating a 12% reduction in net metabolic cost and a 20% improvement in self-reported comfort scores
- Designed, fabricated, and tested lower body perturbation system to simulate trips and slips to enable varied data collection for hip and knee exoskeleton controller testing
- Overhauled hip exoskeleton mechatronics, reducing total system weight by 10% and eliminating external cabling, resulting in a 30% reduction in subject setup time

Mechanical Designer | Plug Power, Concord, MA

November 2023 - April 2025

- Implemented P&I diagram for a stationary 250kW hydrogen fuel cell system by laying out electronic, thermodynamic and mechanical components to allow for ease of access to reference material for all engineers
- Led cross-disciplinary team of in design review meetings to iterate through solutions for product repackaging
- Overhauled system packaging hardware and piping design in PTC Creo for CNC milling and sheet metal manufacturing to achieve 40% and 35% reduction in overall volume and service costs respectively
- Rapidly burned down risk to accelerate project timeline by quickly iterating through schematics and drawings with Microsoft Visio and PTC Creo

Mechanical Engineer | Smartville Inc, Carlsbad, CA

December 2022 - August 2023

- Developed and maintained SolidWorks CAD model and BOM for 1500 kWh battery energy storage system, incorporating DFM and DFA feedback during design refinement stage to drive cost reduction efforts
- Produced machine drawings in accordance to ASME Y14.5 GD&T standards to outsource sheet metal fabrication
- Worked with cross-disciplinary teams to design, fabricate, and operate test fixtures for electrical subsystems
- Prototyped cable harnesses for high and low voltage components in conjunction with electrical engineers
- Coordinated with APAC and domestic suppliers to source electrical components in compliance with UL 1973/74

Mechanical Designer | FORGE LAB, Champaign, IL

July 2021 - August 2022

- Designed LIDAR-based data collection system in SolidWorks to measure atmospheric composition
- Conducted design reviews with NASA Wallops engineers to implement design and manufacturing changes to address concerns about structural stability under rapid vibrations and large temperature deltas

Mechanical Engineering Intern | Instahub, Philadelphia, PA (Remote)

May 2021 - August 2021

- Redesigned PCB enclosures for environmental sensors / automated light switch controllers for all weather use
 - 3D printed and tested prototypes for water intrusion and durability
 - Leveraged Creo Moldflow to simulate injection molding and redesign features to increase manufacturing yields
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SELECTED PROJECTS

ME470 Senior Design Project at UIUC - Expandable optical structure for use in orbital debris detection

- As a team of 6, designed deployable baffle in Fusion360 capable of collapsing to half the operating length
- Performed calculations for desired lens focal length as well as material selections in outer space conditions
- Fabricated prototype for testing using a combination of CNC milling and laser cutting